Pooja Joshi

Data Engineer

jpooja0211@gmail.com
+91 9527820158
Pune, India

Dedicated Data Engineer with over 4.9 years of experience in designing and implementing data processing systems, developing AI-powered document generation solutions, building REST APIs, and optimizing code. Proficient in leveraging technologies such as FastAPI, Pandas, AWS, Docker, and experienced in managing databases using SQL (PostgreSQL, Redshift). Expertise in developing clinical document generation systems, knowledgebase management, and implementing secure multi-tenant data architectures with role-based access control.

TECHNICAL SKILLS

Operating Systems	Windows, Linux	**Programming Languages**	Python, SQL
Tools	Pycharm, Putty, Git, Jira, Dbeaver, CyberDuck, FAE Map, Airflow, PostMan, WinSCP,Grafana	**Framework/ Others**	FastAPI, Pandas, Unit test, AWS, Kafka, Docker, Airflow, PySpark, Redis, Jinja SQL
Databases	SQL (PostgreSQL, Redshift)		

KEY SKILLS

Software Development | Code Optimization and Migration | Debugging | Performance Optimization | Refactoring | RESTful APIs | Continuous Integration/Continuous Deployment (CI/CD) | Data Processing | Quality Assurance | Project Management | Code Review | Unit Testing

WORK EXPERIENE

Software Engineer
Saama Technologies

11/2024 - Present, *Pune, India*
Project: DocGenAI
Achievements/Tasks

- **API Development**: Designed and implemented REST APIs (GET, POST, PUT, DELETE) using Python and Jinja SQL, leveraging Fast API for efficient, high-performance web services.
- **Clinical Document Generation**: Developed solutions to streamline medical writing, enabling faster creation of protocol and CSR documents using AI-driven prompts.
- **Knowledgebase Management**: Enabled users to upload document templates and generate query-based results, improving efficiency in clinical documentation.
- **SQL Query Optimization:** Implemented dynamic SQL query generation for efficient data retrieval and manipulation, ensuring optimized database performance.
- **Data Ingestion Pipelines:** Built robust data ingestion pipelines to process and store large volumes of structured and unstructured data across multiple tenants.
- **Modular Service Development:** Developed modular and reusable services for prompt creation, cloning, and updates, ensuring seamless integration with frontend systems.
- **Security & Access Control:** Ensured data integrity and security by implementing tenant-based isolation and role-based access control (RBAC).
- **Audit Trail Logging:** Automated audit trail logging for all CRUD operations, enabling traceability and compliance with organizational standards.
- **Database Optimization:** Optimized database queries and schema design to handle high-volume transactional data with minimal latency.
- **Cross-functional Collaboration:** Worked closely with cross-functional teams to deliver end-to-end solutions for document generation and prompt execution workflows.
- **Testing & Quality Assurance:** Wrote comprehensive unit and integration test cases to ensure code quality and maintainability, achieving high test coverage across all modules.
- **Project Management:** Applied Agile methodologies, actively contributing to sprint planning, stand-ups, and retrospectives to ensure timely delivery of features.

03/2021 – 11/24, *Pune, India*

Project: SCALE

Achievements/Tasks

- **Adapters Developed:** IBM CD, Oracle Inform EDC, Veeva CTMS, Veeva EDC, REDCap Cloud, Medrio
- **Data Processing and Management:** Designed and implemented robust components for fetching, parsing, and processing raw data from diverse sources using Python, ensuring efficient data handling and accuracy. Developed and maintained adapters to incorporate new requirements, enhancing system flexibility and scalability.
- **Components Engineering:** Engineered components to apply STTM (Source-to-Target Mappings) on raw data, facilitating seamless data transformations.
- **Software Solutions Development:** Write, test, and deploy code using languages such as Python and SQL to develop robust applications tailored to client requirements.
- **Software Development Lifecycle:** Participate in all phases of the software development lifecycle, including requirements gathering, design, coding, testing, deployment, and maintenance.
- **Testing and Quality Assurance:** Conducted comprehensive integration and regression testing on all components with a variety of input sources, ensuring high-quality and reliable software delivery. Additionally, utilized automated testing frameworks and methodologies to streamline the testing process and improve code coverage.
- **Code Optimization and Migration:** Led the migration of existing codebases to more suitable languages and frameworks and applied best practices in object-oriented programming (OOP) and performance optimization to refactor legacy code.
- **Workflow Orchestration:** Implemented and modified Directed Acyclic Graphs (DAGs) in Apache Airflow for efficient task orchestration, ensuring timely and reliable execution of data workflows.
- **Infrastructure and Environment Setup:** Set up and managed environment prerequisites, including the creation of Docker images necessary for the overall processing workflow. Collaborated with DevOps teams to ensure seamless integration and deployment using CI/CD pipelines.
- **Project Management:** Utilized Agile methodologies, participating in sprint planning, daily stand-ups, and retrospectives to ensure timely delivery of features and improvements.

01/2021 - 03/2021, *Pune, India*

Project: IQVIA

Achievements/Tasks

- **Requirement Gathering:** Gather and analyze client requirements, transforming them into effective technical solutions that address user pain points and conduct comprehensive needs assessments and design detailed software solutions, ensuring alignment with business objectives.
- **Data Load Executions and Workflow Development:** Conducted data load executions and developed workflows and implemented and optimized data pipelines.
- **Software Development Lifecycle:** Participate in all phases of the software development lifecycle, including requirements gathering, design, coding, testing, deployment, and maintenance.
- **Adapter Development and Data Integration:** Developed and maintained adapters to efficiently fetch and parse data from diverse sources, ensuring seamless integration and data consistency. Additionally, enhanced existing adapters to support new data formats and sources, improving data retrieval processes.
- **Team Collaboration:** Work closely with cross-functional teams, including engineers, designers, and project managers, to deliver high-quality software products.
- **Technology Exploration:** Tackled challenging tasks involving the latest technologies, staying abreast of industry trends and innovations.
- **Systems Monitoring:** Monitor systems for performance issues, bugs, and errors, promptly addressing any issues to maintain smooth operation.
- **Project Execution and Utility Development:** Participated in ongoing execution activities, ensuring timely and successful project delivery and developed utilities to support project requirements, enhancing overall project efficiency and functionality.

INTERNSHIPS

Persistent Systems (06/2019 - 12/2019)
Pune, India

ACHIEVEMENTS

Saama Shining Star Award

Saama Best Performer Award

Saama High Impact Performer

EDUCATION

Bachelor of Engineering in Information Technology
Savitribai Phule Pune University
2020, *Pune, India*